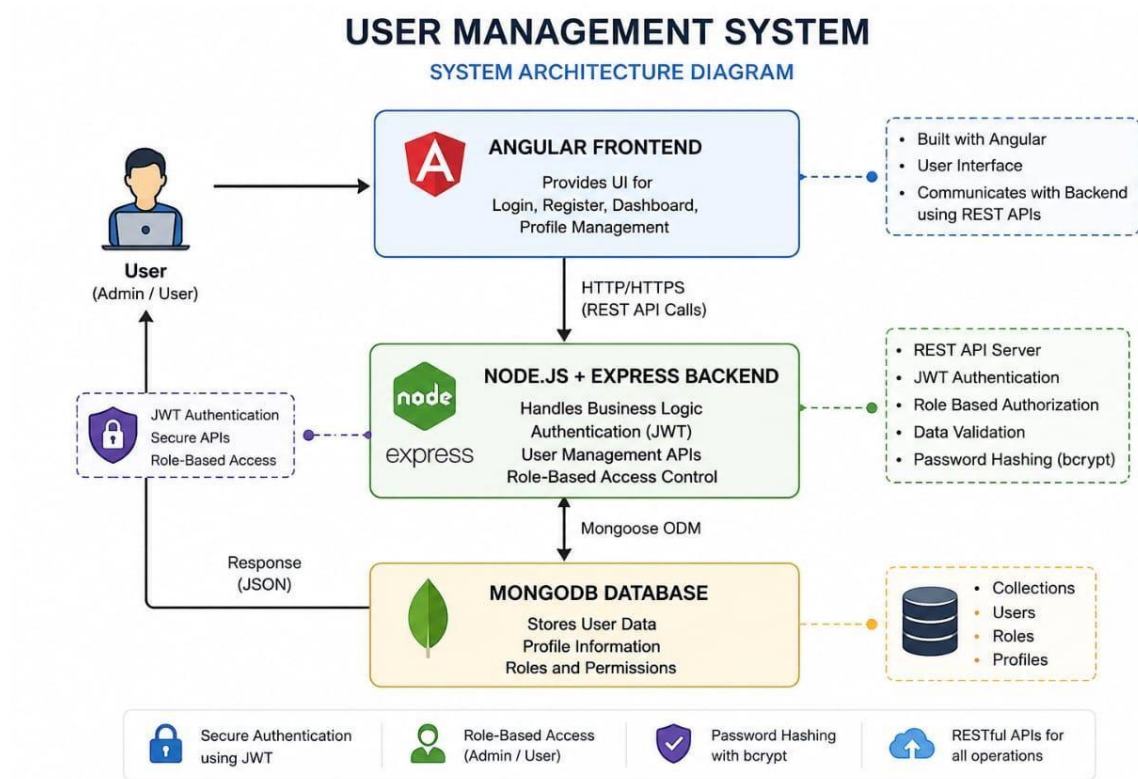


System Architecture



The User Management System follows a three-tier architecture consisting of the presentation layer, application layer, and database layer. The Angular frontend provides an interactive user interface for user registration, login, dashboard access, and profile management. The frontend communicates with the backend through RESTful APIs.

The backend is developed using Node.js and Express.js, which handle user authentication, authorization, business logic, and API

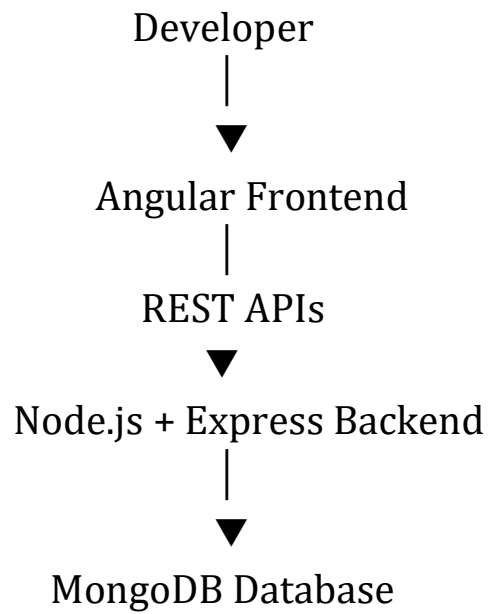
requests. JWT (JSON Web Token) is used for secure user authentication, while bcrypt is used for password hashing and secure password storage.

MongoDB serves as the database layer and stores user information, profile details, roles, and authentication data. Mongoose is used as an Object Data Modeling (ODM) library to facilitate communication between the backend and MongoDB.

This architecture ensures secure authentication, efficient data management, scalability, and seamless communication between the frontend, backend, and database components.

Application Deployment Workflow

The User Management System follows a deployment workflow in which the Angular frontend serves as the user interface and communicates with the Node.js and Express.js backend through REST APIs. The backend processes user requests, performs JWT-based authentication, manages user data, and interacts with the MongoDB database. MongoDB stores user information securely, while bcrypt is used for password encryption. In a deployment environment, the frontend can be hosted on a web server and the backend can be deployed on an application server connected to MongoDB. This workflow ensures secure, reliable, and efficient operation of the application.



The User Management System successfully integrates the Angular frontend, Node.js and Express.js backend, and MongoDB database into a secure and efficient application. The system provides user registration, authentication, role-based access control, and profile management through REST APIs. This architecture ensures reliable communication between components, secure data handling, and scalability for future enhancements.